Huawei Access Node QoS Congestion Event Failure Recovery (KR-ACC-41)

Node: KR-ACC-41 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_2717

Description

The KR-ACC-41 node failure context is attributed to the occurrence of Quality of Service (QoS) congestion events, specifically microbursts, interface tail drops, and QoS queue starvation. This is a critical failure domain that necessitates immediate attention to prevent service disruptions and potential network instability. The Huawei Access node, as a high-capacity platform, relies on sophisticated QoS mechanisms to manage traffic flows and allocate resources efficiently. However, under certain conditions, such as sudden spikes in traffic demand or misconfigured QoS policies, the node may experience congestion, leading to packet loss, increased latency, and ultimately, service outages. The ALARM_2717 alarm code, classified as Major, triggers this SOP to investigate and resolve the underlying issue. A thorough understanding of Huawei's QoS architecture, including the role of the Access node in traffic processing and forwarding, is essential to diagnose and rectify the problem.

Pre-requisite Checks

- Ensure remote hands are dispatched to KR Puram Access for on-site investigation and support.
- Verify the status of redundant links and ensure they are operational.
- Check the node's power supply and ensure it is within acceptable parameters.
- Review the node's environmental conditions, including temperature and humidity.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Insufficient buffer size allocation	Medium
Hardware malfunction or malfunctioning interface	Low

Diagnostic Commands & Expected Outputs

```
Command: display queue statistic, Expected Output: Queue statistics, including queue depth, packet loss, and discard rate., Description: This command provides insight into the current queue status, helping to identify potential congestion points and packet loss areas.

Command: display qos policy, Expected Output: QoS policy configuration, including class maps, policy maps, and service policies., Description: This command displays the configured QoS policies, enabling the identification of potential misconfigurations or inconsistencies.
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Command: display interface detail, Expected Output: Interface details, including statistics, error counters, and operational status., Description: This command provides a comprehensive view of the interface's operational status, helping to identify potential hardware malfunctions or misconfigurations.
```

Resolution Steps

Step 1: Step: Adjust QoS policy configuration, Command: configure qos policy, Description: Update the QoS policy configuration to allocate sufficient buffer size, adjust class maps and policy maps as needed, and apply the revised policy to the affected interfaces.

Step 2: Step: Increase buffer size allocation, Command: buffer-size-adjust, Description: Augment the buffer size allocation to ensure sufficient capacity to handle peak traffic demands and prevent congestion.

Step 3: Step: Replace or repair faulty hardware, Command: replace interface, Description: Replace or repair the malfunctioning interface to restore normal operation and prevent further service disruptions.

Post-Check Verification

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Command: display queue statistic, Expected Output: Queue statistics indicating improved performance and reduced packet loss., Description: Verify that the queue statistics have improved, and packet loss has been mitigated after implementing the resolution steps.

Command: display qos policy, Expected Output: QoS policy configuration indicating the revised policy settings., Description: Confirm that the revised QoS policy configuration has been applied and is operational.

Command: display interface detail, Expected Output: Interface details indicating normal operation and no errors., Description: Verify that the interface is operating normally, with no errors or malfunctions.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: undo configure qos policy, Description: Revert the QoS policy configuration to its original state in case the revised configuration introduced new issues or did not resolve the problem.

- Step: Revert buffer size allocation changes, Command: buffer-size-revert, Description: Revert the buffer size allocation changes to ensure that the original configuration is restored.

- Step: Replace or repair hardware if original hardware is unavailable, Command: replace interface, Description: If the original hardware is not available, replace or repair the faulty hardware to ensure service continuity.